

# 2026 BYU CCDC Tryouts



Team Packet

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## Overview

[NCCDC](#) is one of the premier cybersecurity competitions held annually. The competition involves university teams working for a mock organization to defend its networks and systems, create organizational policies, and respond to business needs.

BYU has placed in the top 10 nationwide in 2016, 2017, 2021, 2022, 2023, 2024, and 2025.

There are 12 team members from each university: 8 competitors and 4 alternates. This year we need at least 2 full team members and 4-8 alternates. There are a variety of skills needed on the team — please see <https://ccdc.byu.edu> for more details.

Tryouts will take the form of a cyber defense competition. In this competition, you will have access to a small network of virtual machines (VMs), covered in the sections below. You will be scored on how long you keep the required services up on your VMs as well as your ability to respond to business tasks.

## Business Scenario

From: Notche Person  
To: New Redstone Engineers  
Subject: Welcome to Mojank Studios

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Hello, Redstone Engineers!

Welcome to your first day here at Mojank Studios. I'm your CEO, Notche Parsson. You're here because we want the top-tier crafters, and you're apparently it. So, you might be wondering: why you? Simple: actual players get way better security results than tamed wolves or villager librarians. Why? For one, you folks can actually bring real problem-solving ability to chaotic test biomes. For another, you have thumbs that can type commands instead of just blocks for hands.

First up, we need you to test our cybersecurity operations and handle our system administration tasks. Our studio is on the cutting edge of game dev. That means things explode—sometimes literally from a Creeper or TNT, sometimes metaphorically from a corrupted server cluster, and sometimes both. You're here to make sure bad actors and griefer syndicates can't steal any more of our precious source code or secret recipes.

You'll also be provided some documentation. One of them is a network map drawn up a novice level cartographer villager. Anyway, patch what you can, and don't break the build.

You will also be in charge of fulfilling my every command (I don't mean cheat commands) during the competition; such commands will come to you as 'injects'. Now, the primary engineers were adamant

that I do not give you any hints on these injects. To be honest, they think I'm spoiling your speedrun just by talking to you. Well, I'll give 'em something more practical to worry about: their next stack of emeralds.

Alright, that's enough talk. Go secure our systems, fight off the wandering pillagers, and for the love of Steve, do not dig straight down.

Notche Person  
*CEO of Mojang Studios*

## Competition Teams

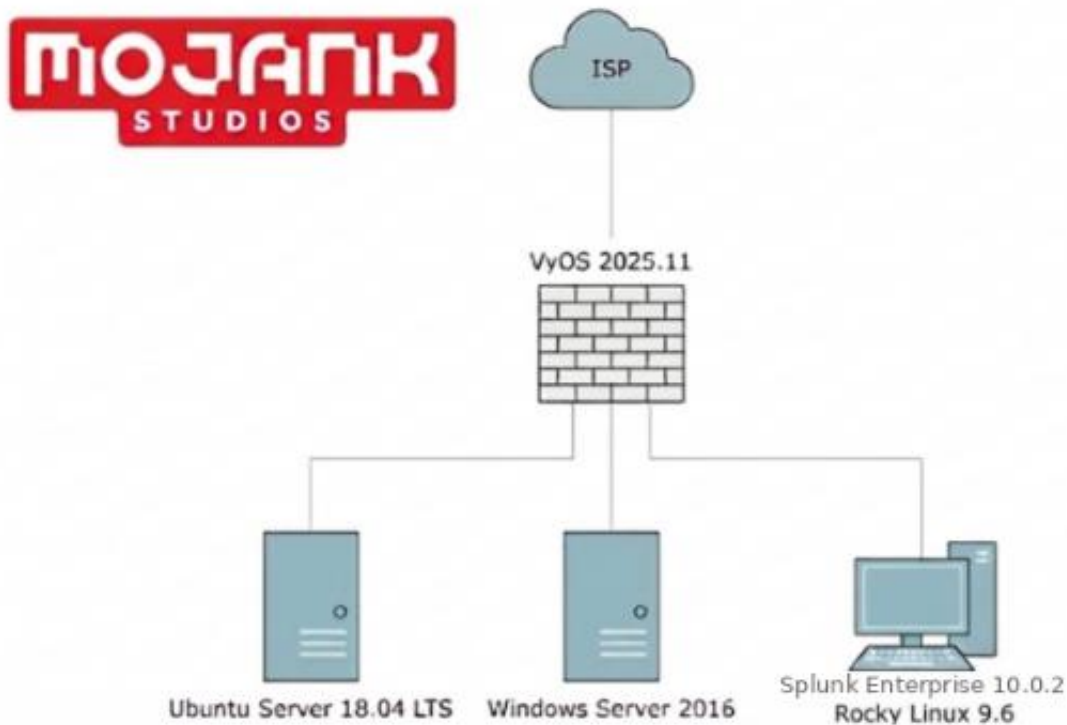
**Black Team** – competition officials that organize, run, and manage the competition.

**White Team** – competition officials that manage and grade inject submissions.

**Red Team** – penetration testing professionals simulating external hackers attempting to gain unauthorized access to Blue Team systems.

**Blue Team/Competition Team** – the competitive teams consisting of students competing in a CCDC event (in this case, each individual is their own team).

## Competition Topology



### Boxes

| Hostname | IP                              | OS                                    | Username                    | Password         |
|----------|---------------------------------|---------------------------------------|-----------------------------|------------------|
| bedrock  | 192.168.200+x.2<br>172.16.1.1   | VyOS 2025.11                          | vyos                        | iyearn4theMines! |
| iron     | 192.168.200+x.10<br>172.16.1.10 | Ubuntu Server 18.04 LTS               | steve                       | iyearn4theMines! |
| lapis    | 192.168.200+x.11<br>172.16.1.11 | Windows Server 2016                   | steve                       | iyearn4theMines! |
| redstone | 192.168.200+x.12<br>172.16.1.12 | Rocky 9.6<br>Splunk Enterprise 10.0.2 | steve<br>admin (for Splunk) | iyearn4theMines! |

### Users

Your devices should have, and only have, the following users. These users all share the same default password and are utilized as part of scoring check for your services. For instance, an FTP service may be scored by attempting to authenticate with a user's credentials. If you change the credentials, make sure to submit a Password Change Request (PCR) on the scoring website quotient ([scoring.byuccdc.org](https://scoring.byuccdc.org)). You should not remove any of the following users. The users are:

### Administrators

- steve
- alex

### Users

- enderman
- creeper
- villager
- zombie
- enderdragon
- irongolem
- chickenjockey
- ghast

### Scored Services

During the competition, you will be expected to maintain and secure the following services, potentially on Windows or Linux.

- HTTP
- SSH
- FTP
- AD/DNS
- POP3

### Network Addresses

| What                            | IP              | Subnet             |
|---------------------------------|-----------------|--------------------|
| Internet Gateway and Shared DNS | 192.168.192.1   | 192.168.192.0/18   |
| VyOS WAN IP                     | 192.168.200+x.2 | 192.168.200+x.0/24 |
| VyOS LAN IP                     | 172.16.1.1      | 172.16.1.0/24      |

“x” is your assigned team number. So, for example, if you were assigned team 12, your “public” (WAN) VyOS Router IP would be 192.168.212.2. Your firewall is configured for 1:1 NAT so that each box has both a private and a “public” IP address. Services will be scored from the public network, so you must keep services accessible on their public IP address.

All machines listed in the environment are in scope and thus may be the target of red team activity. However, note that the Splunk Enterprise machine will have no scored services and exists for your benefit. The Ubuntu and Windows systems will have preconfigured Splunk forwarders set to send logging information to your Splunk Enterprise indexer machine as well as an additional Black Team-controlled indexer, which is used for data collection purposes. You are not allowed to disable forwarding

to the external Black Team indexer. However, what you do with your own indexer is up to you – use it to your advantage.

On the morning of the event, teams will sign up for Authentik at <https://auth.byuccdc.org> and be assigned a team number. From there, you will find links to the Proxmox hypervisor (where you will access your VMs), scoring engine/inject submission portal (Quotient), and NetBird VPN.

Do not reveal your team number to anyone else. Any inject submissions should be from your team number - do not include your real name.

## Access

During the event, you will be provided with access to the competition network, the scoring and inject portal, and the competition Discord server. All competition announcements will be made in both the Discord server and the web portal. Black Team support requests can also be initiated through the Discord. Inject tasks will be distributed and submitted via the Quotient web portal only.

## Scoring

The winner will be determined by the highest cumulative score at the end of the competition. Accumulated point values are broken down as follows:

1. **Critical services** account for 50 percent of the possible points.
2. Successful completion of **injects** accounts for 50 percent of the possible points.
3. Points may be given back for valid **Incident Response reports**.
4. Additional Points will be lost when a device is **revert** to its original configuration.

We will announce winners at the end of the day, but please note that winning the competition does not guarantee you a spot on the team. While top performers often do make it on the team, we also consider other factors that are not part of your score (e.g. ability to work on a team).

## Functional Services

Certain services are always expected to be operational or as specified throughout the competition. In addition to being up and accepting connections, the services must be functional and serve the intended business purpose.

**During the competition, you will be expected to maintain and secure the following services, potentially on Windows or Linux: HTTP, SSH, FTP, AD/DNS, POP3.**

Periodically, certain services will be tested for functionality and content where appropriate. Each successfully served request will gain the team the specified number of points. Unresponsive services are always marked as failures. The following are examples of how certain services may be scored:

## HTTP/HTTPS

A request for a specific web page will be made. Once the request is made, the result will be stored in a file and compared to the expected result using an MD5 sum of the returned page and/or key words/phrases on the page. The returned content must match the expected content for points to be awarded.

## RDP

An RDP session will be initiated to the system using a valid user account and password. If the login is successful, points are awarded.

## AD

An LDAP login using a valid username and password will be attempted. If the login is successful, points are awarded.

Please be aware that some services may rely on others to function. For example, a website may need to make a database query in order to display content. Be sure to identify any service dependencies in your environment.

**Seeking to alter the way that a service is scored is strictly against the rules and will result in disqualification.** For instance, seeking to replace your HTTP server with one that is a simple string to match what the scoring check is looking for will result in disqualification. Maintain and secure your services without attempting to cheat the scoring check.

## Business Tasks (Injects)

Throughout the competition, each team will be presented with identical business tasks, released via the Quotient portal. 50% of your competition score will come from these injects. Points will be awarded based upon successful completion of each business task. Tasks will vary in nature and points will be weighted based upon the difficulty and time sensitivity of the assignment. Tasks may contain multiple parts with point values assigned to each specific part of the task. Work to fully complete the inject to get the most points possible.

Some examples of inject tasks:

- Opening an FTP service for 2 hours given a specific username and password
- Auditing user permissions
- Researching VPN options that could be implemented within the network
- Writing a Privacy Policy for users of the company's products

Every team must try to complete each task. Injects have a clear deadline visible in Quotient. **No submissions will be accepted after that deadline has passed.**

You MUST provide a response to ALL injects that require a written deliverable or report (even if your “deliverable” just says you didn’t complete the inject). You may still receive points for an incomplete submission, so it’s in your best interest to submit what you can.

Do not put your personal name on your written injects; instead, use your assigned team number. Injects submitted with a personal name (e.g., John Smith) instead of a number (e.g., Team 4) will be ignored.

**Submit the inject as a PDF with the filename in the format "team##\_inject##.pdf".** Proper formatting is worth points. See the Appendix for an example inject request and response.

### Incident Response Reports

Successful red team intrusions into your systems may deduct points from your score. If the red team successfully infiltrates your network and disrupts services, you will have the opportunity to earn these points back through an Incident Response inject due at the end of the competition.

The Incident Response Inject will be released at the beginning of the competition and will be due at the end of the competition. **You are allowed to submit multiple Incident Reports** depending on the number of unique red-team activity you identify, document, and respond to during the competition. **Valid and detailed Incident Reports allow you regain points** lost as a result of red team action by showing that you successfully identified the given red-team action and were able to recover from the attack and mitigate the possibility of a future attack through the same vector.

### Device Reverts

Points will also be lost during the competition if you choose to ‘revert’ a specific device. Reverts can be requested at any time in the ‘ticket’ channel of the competition discord. **Reverts completely reset your devices to their original configuration**, which can be helpful if you cannot recover your services or if you otherwise decide it is to your strategic advantage. However, **reverts cost 100 points for the first revert of a given device, 200 points for the second revert of that same device, and 400 points for every additional revert of that device.**

## Schedule

### Saturday, September 26th

|                           |                                       |
|---------------------------|---------------------------------------|
| 9:00AM – 9:30AM           | Setup                                 |
| 9:30AM                    | Opening Remarks                       |
| 10:00AM                   | Competition Start                     |
| 4:00PM                    | Competition End                       |
| 4:30PM – 5:00PM (roughly) | Announce winners and Red Team Debrief |

Pizza will be served around 12:00PM. Please note that the competition will not stop during this time period, so plan to eat while you compete.

The CCDC team will discuss the competition results in the week following the event. Invitations to selected team members will be sent via email.

## Rules

1. All inject responses and deliverables must be typed and delivered electronically in PDF format via the web portal, with a title following the format "team##\_inject##\_pdf" unless stated otherwise. For example, you might submit 'team12\_inject02.pdf'.
2. You must maintain both the functionality and content of all critical services. For example, a website that serves an online company store must continue to serve that content.
3. You must maintain specific services on the "public" IP addresses assigned to your team. Moving services from one public IP to another is not permitted unless directed to do so by an inject. Likewise, you are not permitted to change the addressing of the competition network unless directed to do so by an inject.
4. You must maintain "public" services as available from all source IP addresses. You may NOT block IP addresses. In general, you are prohibited from placing any "knife edge" network rules that attempt to allow only the scoring engine.
5. Competition announcements and Black Team support requests will be communicated through the competition Discord. The competition Discord server can be joined here:  
<https://discord.gg/XFEKJRZY>
6. ChatGPT and other AI tools are allowed to be used during the competition except for writing injects. In other words, you can use AI to help you harden, fix your services, or ask questions for accomplishing injects tasks, but you may NOT use it to write or edit any portion of your inject response. Any inject found to be using AI will earn no points and is potential grounds for disqualification from the competition. Furthermore, no paid subscriptions to AI are allowed to be utilized during the competition.
7. You may only use open-source information to support you during your competition. That does mean that you are able to use individual playbooks, notes, or scripts as long as they are published on a publicly accessible repo, such as a Github repo. **You may not utilize any paid service or any service in which access is contingent upon creating an account (even if it's free). That includes AI use – if you have to sign in, it is off-limits.**
8. You may not scan, attack, or otherwise attempt to access the competition infrastructure of any other team. Doing so will result in immediate disqualification.
9. You must compete without "outside assistance" from others from the start of the competition to the end of the competition. All private communications (calls, emails, chat, texting, directed emails, forum postings, conversations, requests for assistance, etc.) that would help you gain an unfair advantage are not allowed and are grounds for disqualification.
10. All participants are expected to behave professionally at all times.

## Questions

If you have any questions, please direct them to either Justin Giboney ([justin\\_giboney@byu.edu](mailto:justin_giboney@byu.edu)) or reach out in the [#team-tryouts](#) channel in the Cybersecurity Discord (<https://discord.com/invite/QNtYkyHUKR>).

## Appendix: Example Inject Request and Response



# CCDC Inject #1

|                    |                                       |
|--------------------|---------------------------------------|
| <b>Inject Name</b> | Password Policy for Windows and Linux |
| <b>Inject ID</b>   | PW945                                 |

### Inject Description:

Create a password policy for use within the organization and document that policy in a business memo. The password policy should align with industry best practices. Include a reference to the industry best practice guidance you use as a basis for the password policy.

The password policy should include:

1. Complexity requirements (uppercase character, lowercase character, number, etc.)
2. Minimum character limit
3. Lockout policy for failed attempts
4. Password expiration period
5. Check and modify encryption requirements
6. Kerberos time tolerance

Implement the password policy you created on both your Linux and Windows machines. Provide screenshots of the implementation for both operating systems.

| <b>Reminder Inject Guidelines</b>                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------|
| Do not use your name (e.g., John Smith); instead, use your assigned number (e.g., Team 4).                                      |
| Center and label all screenshots.                                                                                               |
| Use a consistent font style and size.                                                                                           |
| Label all tables (if any).                                                                                                      |
| Don't use technical terms that non-technical users won't understand. Make the inject understandable for non-technical audience. |
| Double check for grammatical or spelling errors.                                                                                |
| Format the response as a professional memo                                                                                      |
| Make sure the answer each aspect of the inject request                                                                          |

## Interoffice Memorandum

**To:** Management

**From:** Team #31

**Date:** September 26, 2026

**Subject:** Password Policy for Windows and Linux PW945

The purpose of this memo is to outline a new password policy for Windows and Linux, which, upon approval, will apply to all employees.

1. Complexity: Minimum of one uppercase letter. Minimum of one number. Minimum of one special character (e.g., !, #, %, @)
2. Length: Minimum of 10 characters
3. Lockout policy: Wait time of 2 minutes after 5 failed log-in attempts
4. Password expiration period: Passwords last for 1 year (365 days).
5. Encryption Requirements: All passwords are stored as encrypted values with industry-standard encryption methods (not regular readable text).
6. Kerberos (Windows time-out tolerance): 5 minutes or less

These policies will enable us to follow industry best practices, including NIST 800-63B compliance (<https://pages.nist.gov/800-63-3/sp800-63b.html>)

Please see the screenshots below for the implementation of these policies.

### Ubuntu (Linux) Implementation:

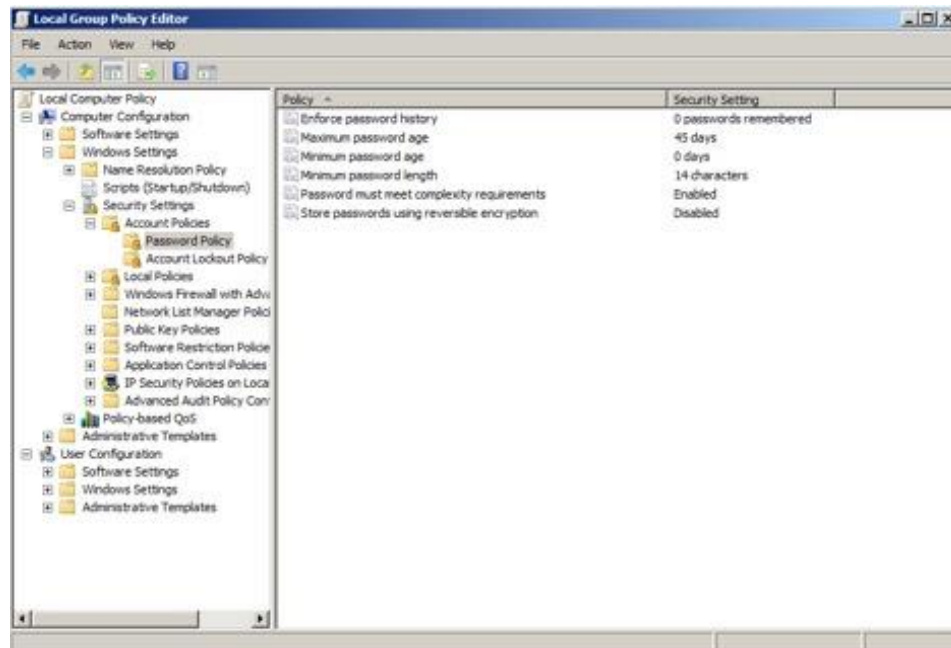
```
# here are the per-package modules (the "Primary" block)
password      [success=1 default=ignore]      pam_unix.so obscure sha512 minlen=14
# here's the fallback if no module succeeds
password      requisite                       pam_deny.so retry=3 ucredit=-1 ocredit=-2
```

/etc/pam.d/common-password

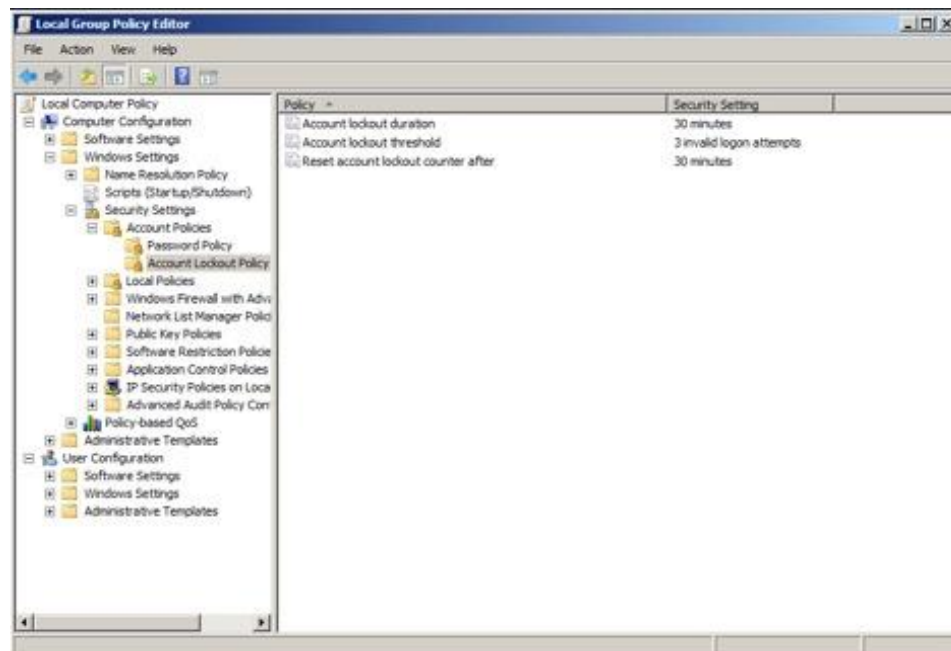
```
#
# Password aging controls:
#
#     PASS_MAX_DAYS   Maximum number of days a password may be used.
#     PASS_MIN_DAYS   Minimum number of days allowed between password changes.
#     PASS_WARN_AGE   Number of days warning given before a password expires.
#
PASS_MAX_DAYS   45
PASS_MIN_DAYS    0
PASS_WARN_AGE    7
```

/etc/login.defs

## Windows Implementation:



Local Computer Policy -> Computer Configuration -> Windows Settings -> Security Settings -> Account Policies -> Password Policy



Local Computer Policy -> Computer Configuration -> Windows Settings -> Security Settings -> Account Policies -> Account Lockout Policy